



Date: 27 / 12 / 2025

Lab Practical #5:

Implementation of Playfair Cipher Technique.

Program:

```
function buildMatrix(key: string): string[][] {
  const k = key.toUpperCase().replace(/J/g, "I").replace(/[^A-Z]/g, "");
  const seen = new Set();
  let chars = "";
  for (const c of k + "ABCDEFGHIJKLMNOPQRSTUVWXYZ") {
    if (!seen.has(c)) {
      seen.add(c);
      chars += c;
    }
  }
  const m: string[][] = [];
  for (let i = 0; i < 5; i++) {
    m.push(chars.slice(i * 5, (i + 1) * 5).split(""));
  }
  return m;
}

function findPos(m: string[][], c: string): [number, number] {
  const ch = c === "J" ? "I" : c;
  for (let i = 0; i < 5; i++) {
    for (let j = 0; j < 5; j++) {
      if (m[i][j] === ch) return [i, j];
    }
  }
  return [0, 0];
}

function prep(s: string): string {
  let r = "";
  s = s.toUpperCase().replace(/J/g, "I").replace(/[^A-Z]/g, "");
  for (let i = 0; i < s.length; i += 2) {
    if (i + 1 < s.length && s[i] === s[i + 1]) {
      r += s[i] + "X";
      i--;
    } else if (i + 1 < s.length) {
      r += s[i] + s[i + 1];
    } else {
      r += s[i] + "X";
    }
  }
  return r;
}
```

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```
function enc(s: string, key: string): string {
  const m = buildMatrix(key);
  const p = prep(s);
  let r = "";
  for (let i = 0; i < p.length; i += 2) {
    const [r1, c1] = findPos(m, p[i]);
    const [r2, c2] = findPos(m, p[i + 1]);
    if (r1 === r2) {
      r += m[r1][(c1 + 1) % 5] + m[r2][(c2 + 1) % 5];
    } else if (c1 === c2) {
      r += m[(r1 + 1) % 5][c1] + m[(r2 + 1) % 5][c2];
    } else {
      r += m[r1][c2] + m[r2][c1];
    }
  }
  return r;
}

function dec(s: string, key: string): string {
  const m = buildMatrix(key);
  const p = prep(s);
  let r = "";
  for (let i = 0; i < p.length; i += 2) {
    const [r1, c1] = findPos(m, p[i]);
    const [r2, c2] = findPos(m, p[i + 1]);
    if (r1 === r2) {
      r += m[r1][(c1 + 4) % 5] + m[r2][(c2 + 4) % 5];
    } else if (c1 === c2) {
      r += m[(r1 + 4) % 5][c1] + m[(r2 + 4) % 5][c2];
    } else {
      r += m[r1][c2] + m[r2][c1];
    }
  }
  return r;
}

const t = "Hello World";
const k = "KEYWORD";
console.log(`Text: ${t}`);
console.log(`Key: ${k}`);
console.log(`Encrypted: ${enc(t, k)}`);
console.log(`Decrypted: ${dec(enc(t, k), k)}`);
```



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Output:

```
~/Desktop/sem-6/ins main*  
> bun playfair-cipher.ts  
Text: Hello World  
Key: KEYWORD  
Encrypted: GYIZSCOKCFBU  
Decrypted: HELXLOWORLDX
```